MODULE 1 · LECTURE 12

Computer-Controlled Devices: Automated Systems for Agri-Input Management

How sensors, controllers and actuators work together; open- and closed-loop control; and how automatic irrigation and fertigation, variable-rate applicators, seed drills and sprayers deliver the right amount of each input.

Learning objectives

After this lecture you should be able to:

1. Describe the parts of a computer-controlled device: sensors, controller, actuators, interface and communication.
2. Distinguish open-loop from closed-loop control and explain on–off (with hysteresis) and PID control.
3. Identify the sensors and controllers used in agricultural automation.
4. Explain automatic irrigation and fertigation systems and calculate run times and injection rates.
5. Explain variable-rate technology and the automatic control of seed drills and sprayers, and calibrate a sprayer.
6. Describe other automated input systems and assess their benefits and limitations.

Scope: Lecture 8 introduced programming, and Lecture 11 computer models. This lecture covers the devices that turn decisions into actions in the field. Calculating crop water and nutrient needs is covered in Lecture 13, the Internet of Things in Lecture 23, precision agriculture in Lecture 26, and AI-based spraying and weeding in Lectures 25 and 27.

1. Why automate the application of inputs?

Agri-inputs are the resources a farmer applies: water, fertiliser, seed, pesticide, feed and energy. Applying them by hand or with uncontrolled machines leads to uneven rates, waste, losses to the environment and heavy labour. A **computer-controlled device** measures conditions, decides what to do according to a program, and acts — delivering the right amount of an input at the right place and time. Precision agriculture applies inputs according to the variation within a field (Zhang et al., 2002), and site-specific application can give equal or higher yields with lower input use and production cost (Balafoutis et al., 2017).

2. Parts of a computer-controlled device

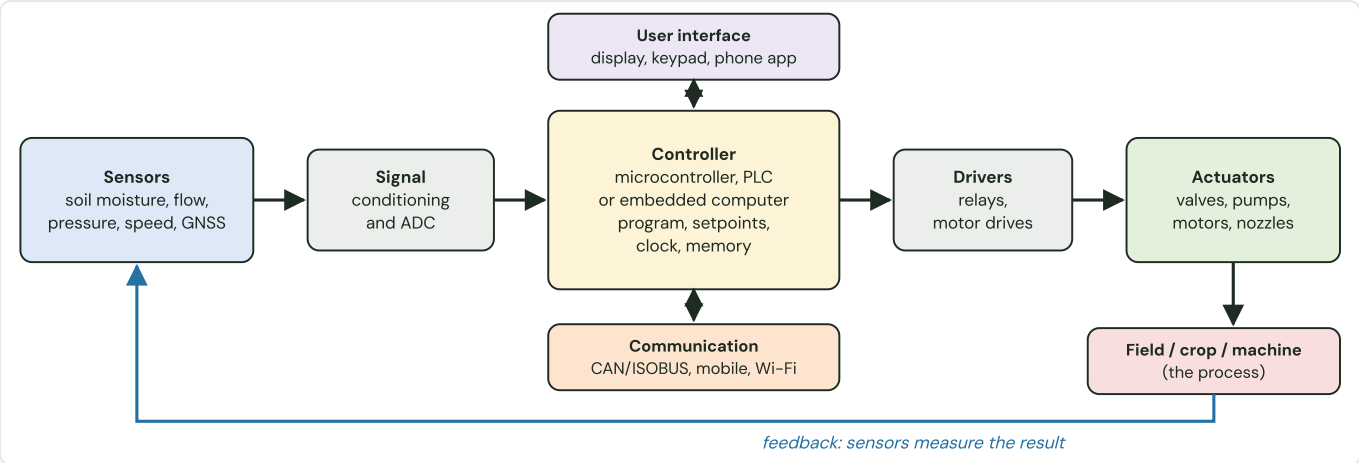


Figure 1. General structure of a computer-controlled device. Original diagram.

Part	Function	Agricultural examples
Sensor	Converts a physical quantity into an electrical signal	Soil-moisture sensor, flow meter, pressure transducer, wheel or radar speed sensor, GNSS receiver
Signal conditioning and ADC	Amplifies and filters the signal; the analogue-to-digital converter turns it into a number	A 10-bit ADC gives readings from 0 to 1023
Controller	Runs the program: compares readings with setpoints and decides on action	Microcontroller board, programmable logic controller (PLC), tractor electronic control unit (ECU)
Driver	Switches or regulates the power needed by the actuator	Relay, contactor, motor driver
Actuator	Converts the control signal into physical action	Solenoid valve, pump, electric or hydraulic motor, nozzle valve
User interface	Lets the operator set targets and see status	Display and keypad, in-cab terminal, smartphone app
Communication	Exchanges data with other devices and software	CAN bus and ISOBUS on tractors; mobile or Wi-Fi links for remote control

3. Control systems

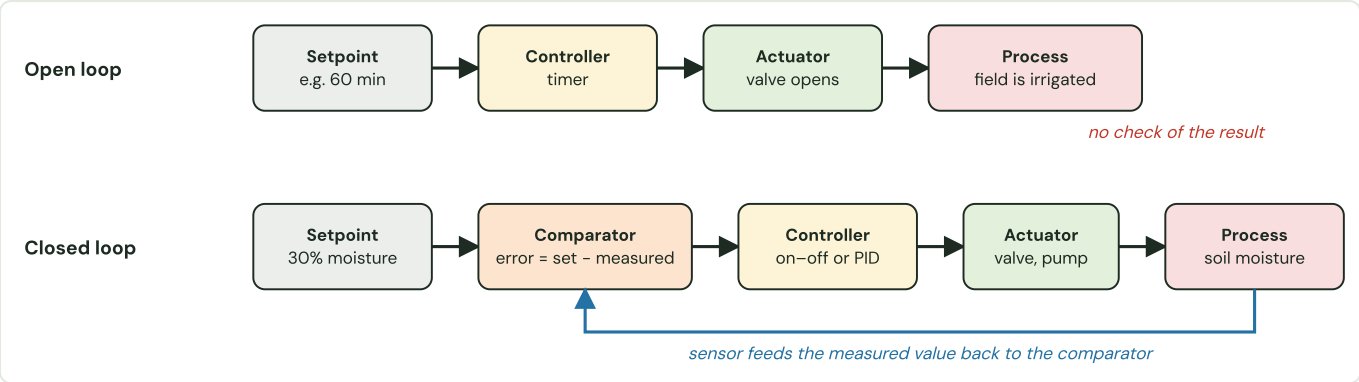


Figure 2. Open-loop and closed-loop (feedback) control of irrigation. Original diagram.

- **Open-loop control** acts without measuring the result — e.g. a timer that opens a valve for 60 minutes every morning, whether or not it has rained.
- **Closed-loop (feedback) control** measures the output, compares it with the **setpoint**, and corrects the difference (**error**) — e.g. irrigating only when soil moisture is below the target. Feedback makes the system respond to disturbances such as rain or hot weather (Ogata, 2010).

Control strategy	How it works	Typical use
Time-based (timer)	Fixed start time and duration	Simple drip and sprinkler controllers
On–off	Actuator fully on or fully off depending on the measured value	Pump switched by a soil-moisture or tank-level sensor
On–off with hysteresis (deadband)	Separate switch-on and switch-off thresholds stop the actuator from switching rapidly on and off	Irrigation between 25% and 35% moisture; greenhouse heater
Proportional (P)	Output proportional to the error	Opening a mixing valve in proportion to the EC error
PID	Proportional + integral + derivative action — removes steady error and damps oscillation	Sprayer pressure and flow, greenhouse temperature, drive speeds

$$\text{Error } e(t) = \text{setpoint} - \text{measured value} \qquad u(t) = Kp \cdot e(t) + Ki \cdot \int e(t)dt + Kd \cdot de(t)/dt$$

In the PID equation (Ogata, 2010), the **proportional** term reacts to the present error, the **integral** term to the error accumulated over time, and the **derivative** term to how fast the error is changing; the gains Kp, Ki and Kd are tuned for each system.

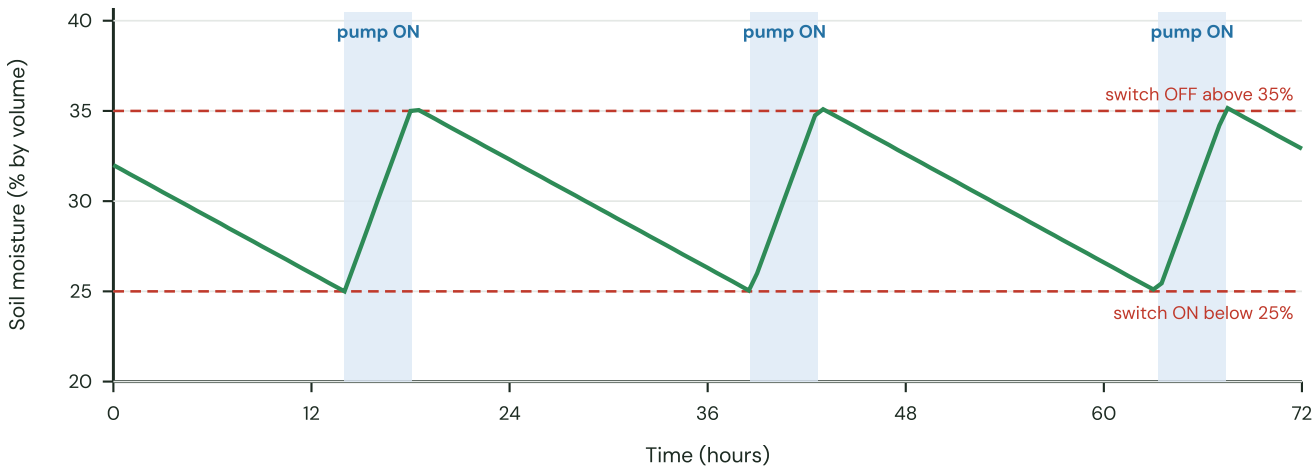


Figure 3. Simulated soil moisture under on–off irrigation control with hysteresis (pump on below 25%, off above 35%). Original chart from an illustrative simulation.

Example program — on–off irrigation control with hysteresis

```
// Soil-moisture irrigation controller with hysteresis (Arduino-style C++, illustrative)
const int SENSOR_PIN = A0;        // analogue soil-moisture sensor
const int PUMP_PIN = 7;           // relay that switches the pump
const float ON_BELOW = 25.0;      // % moisture: start the pump below this
const float OFF_ABOVE = 35.0;     // % moisture: stop the pump above this
bool pumpOn = false;

void setup() {
  pinMode(PUMP_PIN, OUTPUT);
  digitalWrite(PUMP_PIN, LOW);
}

void loop() {
  int raw = analogRead(SENSOR_PIN);                    // 0 to 1023
  float moisture = (850 - raw) * 50.0 / (850 - 400);    // calibration: dry 850 = 0 %, wet 400 = 50 %
  if (!pumpOn && moisture < ON_BELOW) {
    pumpOn = true;
  } else if (pumpOn && moisture > OFF_ABOVE) {
    pumpOn = false;
  }
  digitalWrite(PUMP_PIN, pumpOn ? HIGH : LOW);
  delay(60000);                                        // check once a minute
}
```

This listing is illustrative and was not compiled for a board. Its switching logic was tested in a simulation (Figure 3), and the sensor calibration values must be found for each sensor and soil. With a single threshold instead of two, the pump would switch on and off every few minutes near 25%, wearing out the relay and pump.

4. Sensors and controllers used in agriculture

4.1 Measuring soil moisture

Method	Principle	Notes
Gravimetric (oven-dry)	Weigh a soil sample, dry it at 105 °C and weigh again	The reference method; slow and destructive, used to calibrate sensors
Tensiometer	Measures soil-water tension (suction) through a porous cup	Shows how hard roots must work to extract water; simple, works in wetter soils
Time-domain reflectometry (TDR)	Travel time of an electromagnetic pulse depends on the soil's dielectric permittivity, which water dominates	Accurate; costlier
Capacitance / frequency-domain sensors	Capacitance of the soil around the probe changes with water content	Low cost, widely used in automatic controllers; need soil-specific calibration
Resistance (gypsum) blocks	Electrical resistance of a block in equilibrium with the soil	Cheap; affected by salinity

Soil water content governs infiltration, plant uptake and nutrient availability, so measuring it accurately is central to automatic irrigation and fertigation; the main methods and their physical principles are reviewed by Bittelli (2011).

4.2 Controllers

Controller	Features	Where used
Microcontroller board	A small, low-cost computer on one chip with input/output pins; programmed in C/C++; some have built-in Wi-Fi	Prototype and low-cost irrigation controllers, data loggers, student projects
Programmable logic controller (PLC)	Rugged industrial controller with many inputs and outputs; programmed in ladder logic and other standard languages; highly reliable	Large irrigation pump stations, greenhouses, dairy and feed plants, grain dryers
Embedded / single-board computer	Runs a full operating system; handles cameras, networking and complex software	Gateways, image-based systems, local dashboards
Dedicated irrigation controller	Ready-made unit with zones, schedules and sensor inputs	Farms, orchards, landscapes
Tractor ECU and task controller	Controls implement functions and rates; communicates over ISOBUS	Seed drills, sprayers, fertiliser spreaders

5. Automatic irrigation and fertigation

Automatic micro-irrigation controllers open and close **solenoid valves** zone by zone on a schedule or in response to soil-moisture, rain or weather data, and switch the pump. In India, the **Per Drop More Crop** component of the Pradhan Mantri Krishi Sinchayee Yojana supports drip and sprinkler systems. It began in 2006 as a micro-irrigation scheme and had brought more than 115 lakh hectares under micro-irrigation by July 2026, with 55% assistance for small and marginal farmers and 45% for others, for up to five hectares per beneficiary (News On AIR, 2026).

Fertigation is the application of fertiliser dissolved in irrigation water. It feeds the crop little and often, close to the roots, and is easy to automate (Kafkafi & Tarchitzky, 2011).

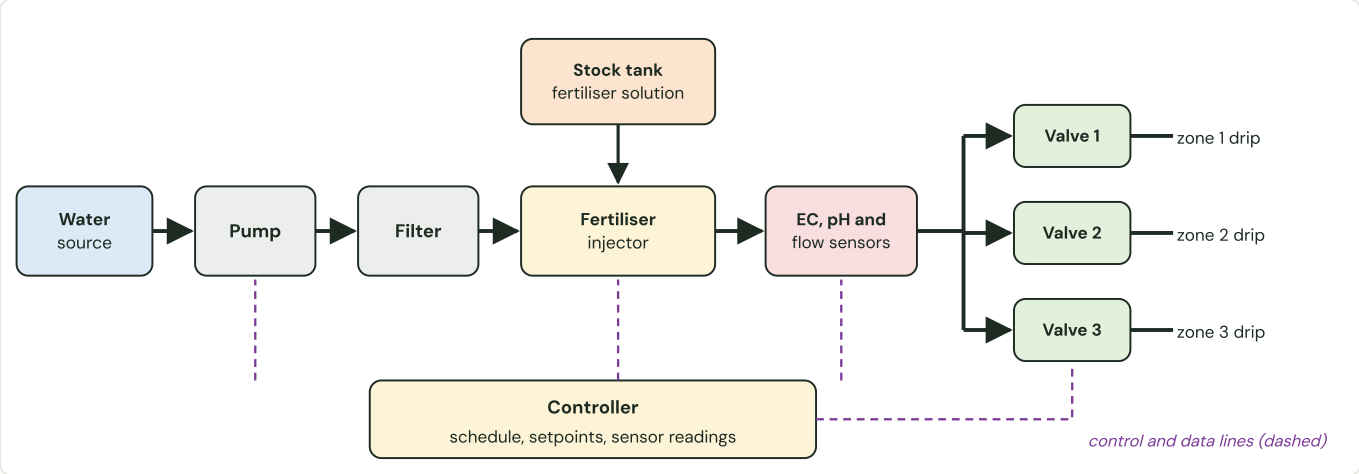


Figure 4. Layout of an automated drip irrigation and fertigation system with three zones. Original diagram.

Fertiliser injector	Principle	Features
Venturi injector	A constriction in the pipe speeds up the water and lowers its pressure, sucking fertiliser solution from the tank	No moving parts and no power needed; causes some pressure loss
Pressure-differential (by-pass) tank	A throttle valve diverts part of the flow through a closed tank containing fertiliser	Simple; can take solid fertiliser, but the concentration falls as the fertiliser dissolves
Injection (dosing) pump	A piston or diaphragm pump, driven hydraulically or electrically, injects solution at a set ratio	Accurate, constant concentration, easy to automate; most common with controllers

Fertiliser can be dosed **quantitatively** (a measured amount in each irrigation) or **proportionally** (a constant ratio of solution to water, giving a constant nutrient concentration) (Kafkafi & Tarchitzky, 2011). Automated systems check the result with **electrical conductivity (EC)** and **pH** sensors.

Worked example 1 — drip run time and zone sequencing (illustrative values)
Each plant needs 4 L of water a day and has 2 emitters of 2 L/h → run time = 4 ÷ (2 × 2) = 1 hour.
A 1-ha block with 5,000 plants has 10,000 emitters → flow = 10,000 × 2 = 20,000 L/h = 20 m³/h (5.56 L/s).
If the pump can supply only 10 m³/h, the controller divides the block into two zones and runs them one after the other, 1 hour each.

Worked example 2 — fertigation injection rate (illustrative values)
Apply 2 kg N per hectare per day as urea (46% N): urea = 2 ÷ 0.46 = 4.35 kg, dissolved in 50 L of water (87 g/L).
To inject the 50 L over 40 minutes of the irrigation cycle, the dosing pump is set to 50 ÷ 40 = 1.25 L/min.

6. Variable-rate technology, seed drills and sprayers

Variable-rate technology (VRT) changes the application rate as the machine moves across the field, so that each part of the field receives what it needs.

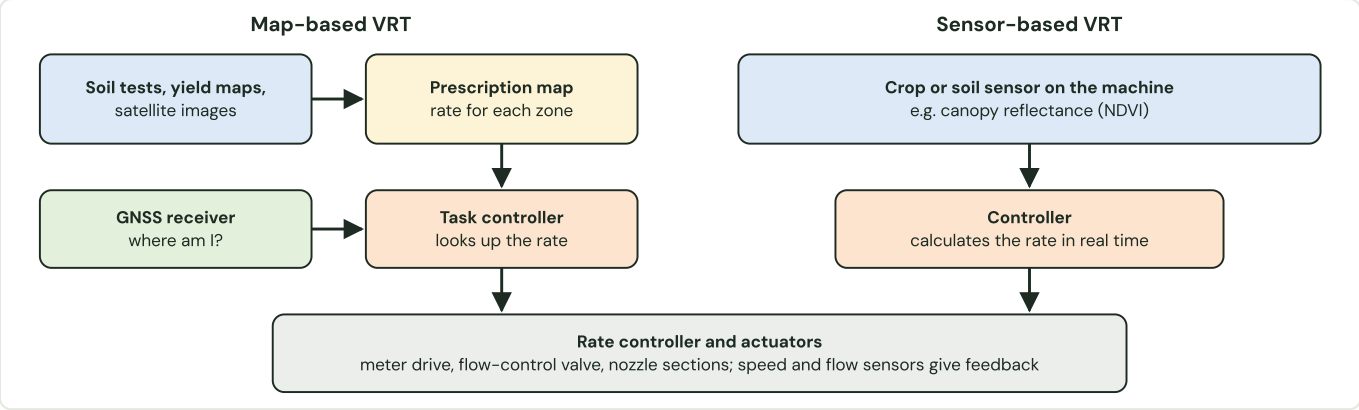


Figure 5. Map-based and sensor-based variable-rate application. Original diagram.

- **Map-based VRT** — a **prescription map** made from soil tests, yield maps or satellite images is loaded into the task controller; a **GNSS** receiver tells it where the machine is, and it sets the rate for that zone.
- **Sensor-based VRT** — sensors on the machine measure the crop or soil (e.g. canopy reflectance for nitrogen) and the rate is calculated in real time, without a prior map.
- In both, a **rate controller** drives the metering mechanism or flow-control valve and uses speed and flow sensors as feedback.

Worked example 3 — nitrogen saved by VRT (illustrative values)
Uniform application on a 10-ha field: 10 × 120 = 1,200 kg N.
Prescription: 4 ha at 80, 4 ha at 110 and 2 ha at 140 kg N/ha → 320 + 440 + 280 = 1,040 kg N.
Saving = 160 kg N (13.3%) = 160 ÷ 0.46 ≈ 348 kg of urea (about 7.7 bags of 45 kg). VRT may also raise the rate where the crop needs more, so its main gain is putting nutrients where they are used.

Seed drills and planters

- **Electronic seed monitors** count seeds in each tube and warn of blockages or gaps.
- **Electric-drive seed meters** replace ground-wheel drives, so seed rate can be changed from the cab or by a prescription map.
- **Section control** switches off rows or sections over areas already sown or outside the field boundary, avoiding overlaps.

Sprayers

For a boom sprayer the application rate depends on nozzle flow, nozzle spacing and forward speed. The constant 600 converts L/min, km/h and metres into L/ha:

$$\text{Rate (L/ha)} = \text{nozzle flow (L/min)} \times 600 \div [\text{speed (km/h)} \times \text{nozzle spacing (m)}] \qquad \text{Speed (km/h)} = \text{distance (m)} \times 3.6 \div \text{time (s)}$$

Worked example 4 — sprayer calibration and rate control (illustrative values)
Target 300 L/ha, nozzles 0.5 m apart on a 12-m boom (24 nozzles). Driving 100 m takes 60 s → speed = 100 × 3.6 ÷ 60 = 6 km/h.
Required nozzle flow = 300 × 6 × 0.5 ÷ 600 = 1.5 L/min; boom flow = 24 × 1.5 = 36 L/min.
If the tractor speeds up to 7.5 km/h, the rate would fall to 240 L/ha unless flow increases. An automatic **rate controller** raises it to 1.875 L/min per nozzle (45 L/min).
Because nozzle flow varies with the square root of pressure, this needs $(1.875 \div 1.5)^2 = 1.56$ times the pressure — 3 bar becomes about 4.7 bar, which may change droplet size. For large speed changes it is better to change nozzles or use pulse-width-modulated nozzle control.

The speed check and nozzle-flow calculation follow a manufacturer's field-sprayer guide (HARDI, n.d.). Before spraying, also check each nozzle's actual output by collecting it for one minute.

- **Automatic section control** uses GNSS to switch boom sections off over areas already sprayed, headlands and field edges.
- **Pulse-width modulation (PWM)** pulses each nozzle's solenoid valve rapidly on and off, controlling flow without changing pressure — allowing individual nozzle control.
- **ISOBUS** — the ISO 11783 standard defines communication between tractors, implements and farm software, so a terminal from one maker can control an implement from another (Agricultural Industry Electronics Foundation [AEF], n.d.). This allows one in-cab terminal to run sprayers, seed drills and spreaders.

7. Other automated input systems

System	What is automated	Notes
Drone (UAV) sprayers	Pre-planned flight paths, height and swath; flow controller linked to flying speed	Used for pesticide and liquid-fertiliser spraying, especially in tall or wet crops; India supports drone services through women self-help groups. Covered further in Lecture 25
Greenhouse and polyhouse control	Temperature, humidity, CO_2, ventilation fans, foggers, shade screens and fertigation	Closed-loop control with several sensors; often PLC-based
Livestock feeding and watering	Feed dispensers that recognise animals by RFID tags and give each its ration; automatic waterers	See Lecture 24 for AI in livestock management
Tractor guidance and auto-steer	GNSS-based steering along straight or curved lines	Reduces overlaps and skips when applying inputs

8. Benefits, limitations and good practice

Benefits	Limitations
Accurate, uniform rates; less waste of water, fertiliser and chemicals	Purchase cost; small farms may need shared or hired equipment
Labour saving and timely operation	Needs reliable power, maintenance and spare parts
Responds to changing conditions (closed loop)	Sensors drift and must be calibrated for each soil or crop
Lower losses to groundwater and air	Skills needed for set-up, programming and repair
Records of what was applied, where and when	A faulty sensor or program can over- or under-apply on a large area

Good practice: calibrate sensors and sprayers at the start of each season; flush and clean drip filters and injectors; protect electronics from water, heat and rodents; fit manual overrides and fail-safe defaults (pump off if a sensor fails); and keep records. Small farmers can access automated equipment through custom-hiring centres, FPOs and service providers.

9. Key takeaways

- A computer-controlled device links sensors, a controller and actuators, with an interface and communication links.
- Open-loop control acts without checking the result; closed-loop control uses feedback to hold a setpoint.
- On–off control needs hysteresis to avoid rapid switching; PID control combines proportional, integral and derivative action.
- Microcontrollers, PLCs, embedded computers and tractor ECUs are the main controllers in agriculture.
- Automatic irrigation and fertigation use solenoid valves, injectors (Venturi, by-pass tank, dosing pump) and EC/pH sensors.
- Variable-rate technology is map-based or sensor-based; rate controllers, section control, PWM nozzles and ISOBUS automate seed drills and sprayers.
- Sprayer rate = nozzle flow × 600 ÷ (speed × spacing); calibration and maintenance are essential for accurate automation.

10. Examination questions

Short-answer questions (2 marks)

Q1. Define sensor and actuator, with one agricultural example of each.

A sensor converts a physical quantity into an electrical signal (soil-moisture sensor); an actuator converts a control signal into action (solenoid valve).

Q2. Distinguish open-loop and closed-loop control.

Open-loop acts without measuring the result (timer); closed-loop uses feedback to correct the error (soil-moisture-based irrigation).

Q3. Why is hysteresis used in on–off control?

Separate on and off thresholds prevent rapid switching that wears out pumps and relays.

Q4. Write the PID control equation.

$u(t) = K_p \cdot e(t) + K_i \cdot \int e(t)dt + K_d \cdot de(t)/dt$.

Q5. What is a PLC?

A programmable logic controller — a rugged industrial controller with many inputs and outputs, programmed in ladder logic and similar languages.

Q6. Name three types of fertiliser injectors.

Venturi injector, pressure-differential (by-pass) tank, injection (dosing) pump.

Q7. Define fertigation.

Application of fertiliser dissolved in irrigation water.

Q8. Distinguish map-based and sensor-based VRT.

Map-based uses a prescription map and GNSS position; sensor-based measures the crop or soil on the go and sets the rate in real time.

Q9. What is ISOBUS?

The ISO 11783 standard for communication between tractors, implements and farm software.

Q10. A nozzle delivers 1.5 L/min at 6 km/h with 0.5 m spacing. Find the application rate.

$1.5 \times 600 \div (6 \times 0.5) = 300$ L/ha.

Long-answer questions (5–10 marks) — answer outlines

Q1. Describe the components of a computer-controlled device with a block diagram.

Figure 1; sensors, signal conditioning and ADC, controller, drivers, actuators, interface, communication; feedback; examples.

Q2. Explain open-loop and closed-loop control. Describe on–off control with hysteresis and PID control.

Figure 2; control strategy table; PID equation; Figure 3 and the example program.

Q3. Explain the methods of measuring soil moisture used in automatic irrigation.

Gravimetric, tensiometer, TDR, capacitance, resistance blocks — principles and notes; calibration.

Q4. Describe an automated drip irrigation and fertigation system.

Figure 4; controller, valves, sensors; injector types; quantitative and proportional dosing; worked examples 1 and 2; Per Drop More Crop.

Q5. Explain variable-rate technology and its use in fertiliser application.

Figure 5; map-based vs sensor-based; prescription map, GNSS, rate controller; worked example 3.

Q6. Explain how seed drills and sprayers are automatically controlled. Calibrate a boom sprayer for a given rate.

Seed monitors, electric meters, section control; rate controller, PWM, ISOBUS; formula and worked example 4.

11. Glossary

Term	Meaning
Actuator	A device that converts a control signal into physical action
ADC	Analogue-to-digital converter
Closed-loop control	Control that uses feedback from the measured output
Controller	The device that runs the control program
ECU	Electronic control unit on a tractor or machine
EC	Electrical conductivity — indicates the salt or nutrient concentration of a solution
Fertigation	Applying fertiliser through irrigation water
GNSS	Global navigation satellite system, such as GPS
Hysteresis (deadband)	A gap between switch-on and switch-off values
ISOBUS	ISO 11783 standard for tractor–implement communication
Microcontroller	A small computer on a single chip with input/output pins
Open-loop control	Control without feedback
PID	Proportional–integral–derivative control
PLC	Programmable logic controller
Prescription map	A map of the input rate to be applied in each part of a field
PWM	Pulse-width modulation — rapid on/off pulsing to control average flow or power
Rate controller	Controller that keeps the application rate on target as speed changes
Relay	An electrically operated switch
Section control	Automatic switching of implement sections to avoid overlaps
Setpoint	The target value a control system tries to maintain
Solenoid valve	A valve opened and closed by an electromagnet
VRT	Variable-rate technology

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All diagrams, the chart and the program listing are original. Standard material on agricultural machines and their control follows Srivastava et al. (2006) and Ogata (2010). Values in the worked examples are illustrative, and every result shown was recalculated.

Practice MCQs

Q1. A device that converts a physical quantity into an electrical signal is a

(a) sensor (b) actuator
(c) relay (d) controller

Q2. In an automatic drip system, the solenoid valve acts as

(a) a sensor (b) an actuator
(c) a controller (d) a filter

Q3. A timer that opens a valve for 60 minutes every day, regardless of rain, is an example of

(a) closed-loop control (b) open-loop control
(c) PID control (d) feedback control

Q4. Separate switch-on and switch-off thresholds in on–off control are used to

(a) increase pump speed (b) prevent rapid on–off switching
(c) measure soil moisture (d) save memory

Q5. In PID control, the integral term responds to

(a) the present error only (b) the error accumulated over time
(c) how fast the error changes (d) the setpoint only

Q6. A rugged industrial controller programmed in ladder logic is a

(a) PLC (b) GNSS receiver
(c) tensiometer (d) relay

Q7. The reference (standard) method for measuring soil water content is

(a) TDR (b) gravimetric oven-drying
(c) capacitance probe (d) gypsum block

Q8. A fertiliser injector that uses a constriction to create suction is a

(a) dosing pump (b) Venturi injector
(c) by-pass tank (d) solenoid valve

Q9. A nozzle delivering 1.5 L/min at 6 km/h with 0.5 m spacing applies

(a) 150 L/ha (b) 300 L/ha
(c) 450 L/ha (d) 600 L/ha

Q10. Variable-rate application that uses a prescription map and GNSS position is

(a) sensor-based VRT (b) map-based VRT
(c) open-loop timing (d) section control

Q11. The standard for communication between tractors and implements is

(a) ISO 9001 (b) ISO 11783 (ISOBUS)

(c) IEEE 802.11 (d) ISO 14001

Q12. Under Per Drop More Crop, small and marginal farmers receive financial assistance of

(a) 25% (b) 45%

(c) 55% (d) 100%

Answers

Q1 — (a) sensor

Q2 — (b) an actuator

Q3 — (b) open-loop control

Q4 — (b) prevent rapid on–off switching

Q5 — (b) the error accumulated over time

Q6 — (a) PLC

Q7 — (b) gravimetric oven-drying

Q8 — (b) Venturi injector

Q9 — (b) 300 L/ha

Q10 — (b) map-based VRT

Q11 — (b) ISO 11783 (ISOBUS)

Q12 — (c) 55%